📊 Supply Chain & Financial Analytics

📌 Project Overview

This project presents a **Power BI dashboard** that delivers a unified view of business performance across financial and operational domains. It enables stakeholders to monitor key metrics, identify trends, and make **data-driven decisions** efficiently.

🎯 Business Objective

- Monitor revenue and profitability trends
- Evaluate supply chain and supplier performance
- Identify inventory shortages and risks
- Track shipment efficiency and delivery performance
- Analyze customer behavior and contribution

📂 Data Source

The dashboard is built using multiple structured CSV datasets:

- Sales
- Customer
- Product
- Supplier
- Inventory
- Procurement
- Production
- Shipment
- Facility
- Date Table

🧹 Data Preparation

- Performed data cleaning and transformation using **Power Query**
- Designed a **Star Schema data model** for efficient analysis
- Created a centralized **Date Table** for time-based reporting
- Handled missing values and ensured data consistenc

📊 Key KPIs

- Total Revenue
- Gross Revenue
- Profit
- Profit Margin %
- Total Orders
- Inventory Quantity
- Shipment Quantity
- On-Time Delivery %
- Total Customers

📄 Dashboard Pages

1️⃣ Index Page

Serves as a navigation hub to access all dashboard sections.

2️⃣ Executive Overview

Provides a high-level summary of business performance using key KPIs and revenue trends.

3️⃣ Supply Analysis

Analyzes supplier performance, procurement costs, and contribution to overall operations.

4️⃣ Inventory Analysis

Tracks stock levels across products and locations, highlighting low inventory risks.

5️⃣ Shipment Analysis

Evaluates shipment trends, delivery performance, and identifies delay patterns.

6️⃣ Customer Analysis

Explores customer segmentation, top customers, and purchasing behavior.

🎛️ Filters & Interactivity

- Date-based filtering
- Product category selection
- Customer and supplier filters
- Cross-filtering enabled across all visuals for dynamic analysis

🧮 DAX Measures

Total Revenue = SUM(Sales[net_revenue])

Gross Revenue = SUM(Sales[gross_revenue])

Profit = SUM(Sales[profit])

Profit Margin % = DIVIDE([Profit], [Total Revenue], 0)

Total Orders = DISTINCTCOUNT(Sales[order_number])

Avg Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

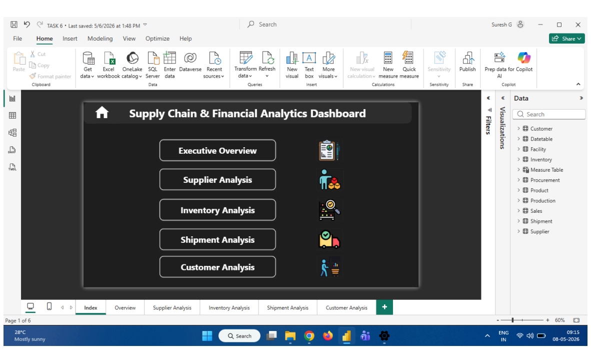
📈 Key Insights

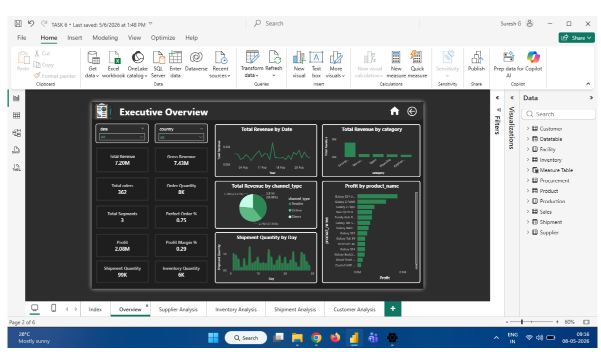
- Revenue trends provide visibility into overall business growth
- Supplier analysis highlights high-cost and high-performing vendors
- Inventory analysis helps identify potential stock shortages
- Shipment analysis reveals delivery efficiency and delay issues
- Customer analysis identifies key contributors to revenue

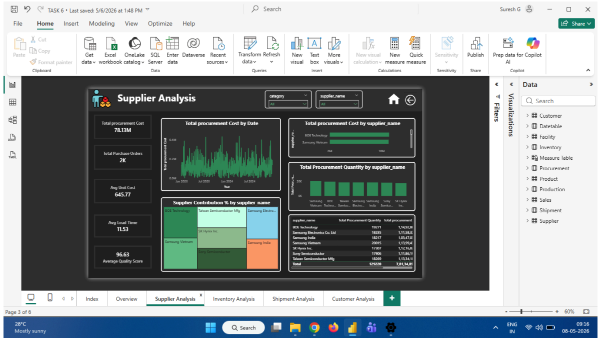
🧾 Conclusion

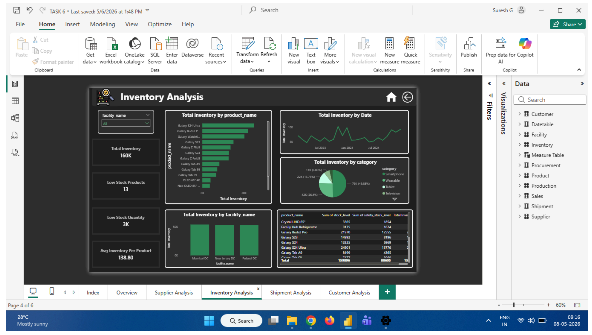
This dashboard provides a **comprehensive and interactive view of business performance**, enabling stakeholders to make informed decisions across financial, operational, and customer domains.

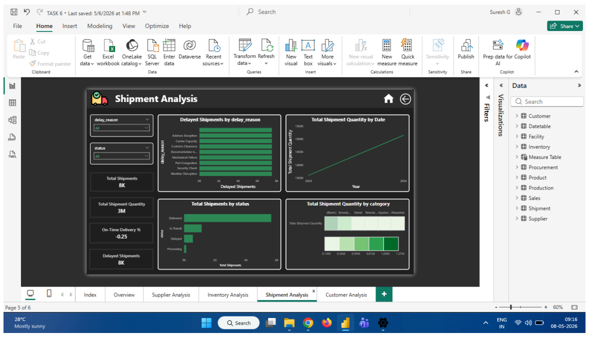
📷 Screenshots

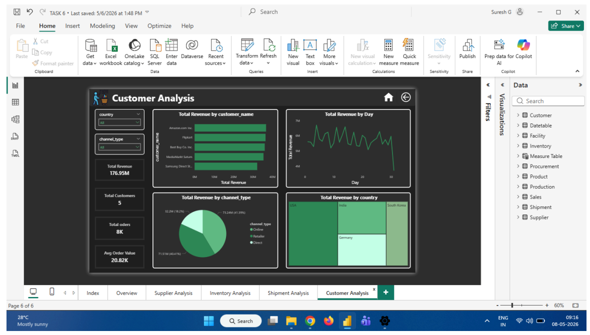





Supplier Analysis


Inventory Analysis


Shipment Analysis


Customer Analysis